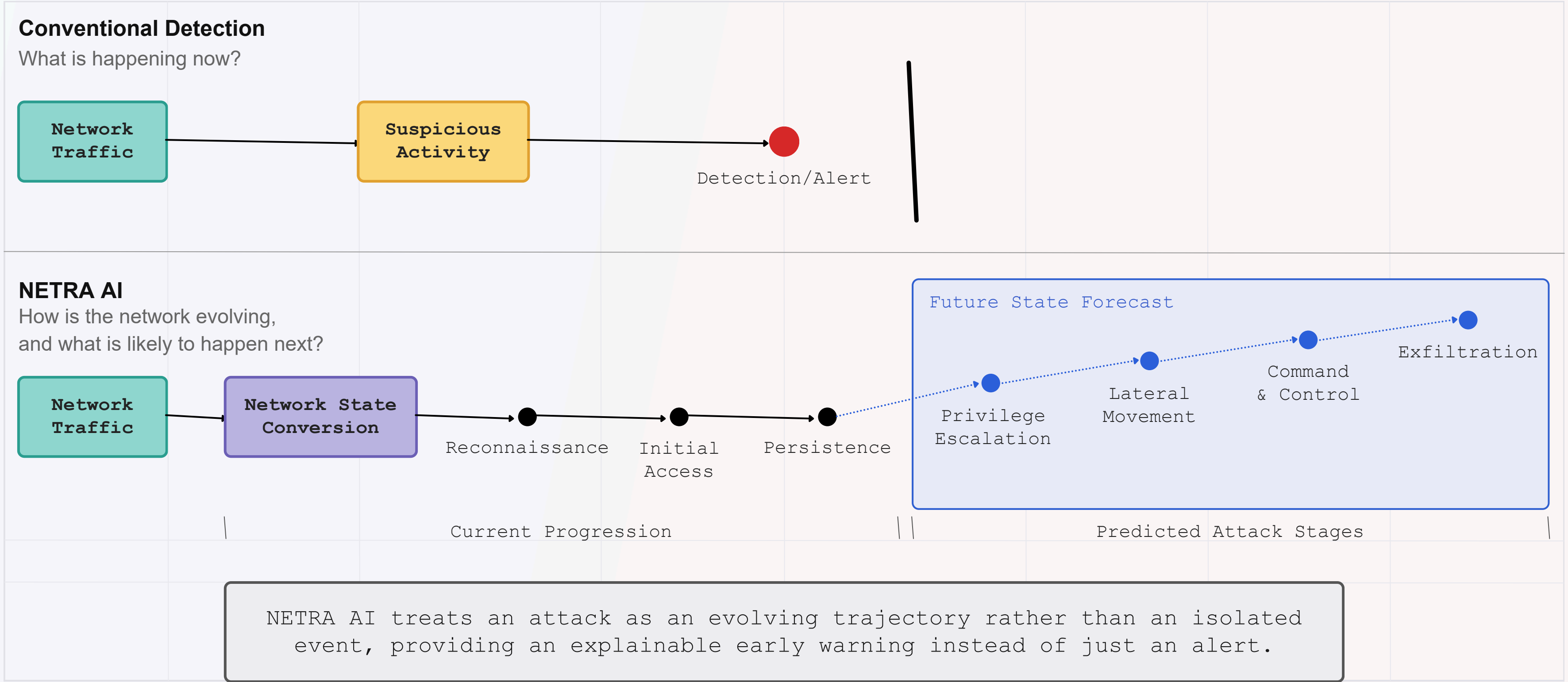
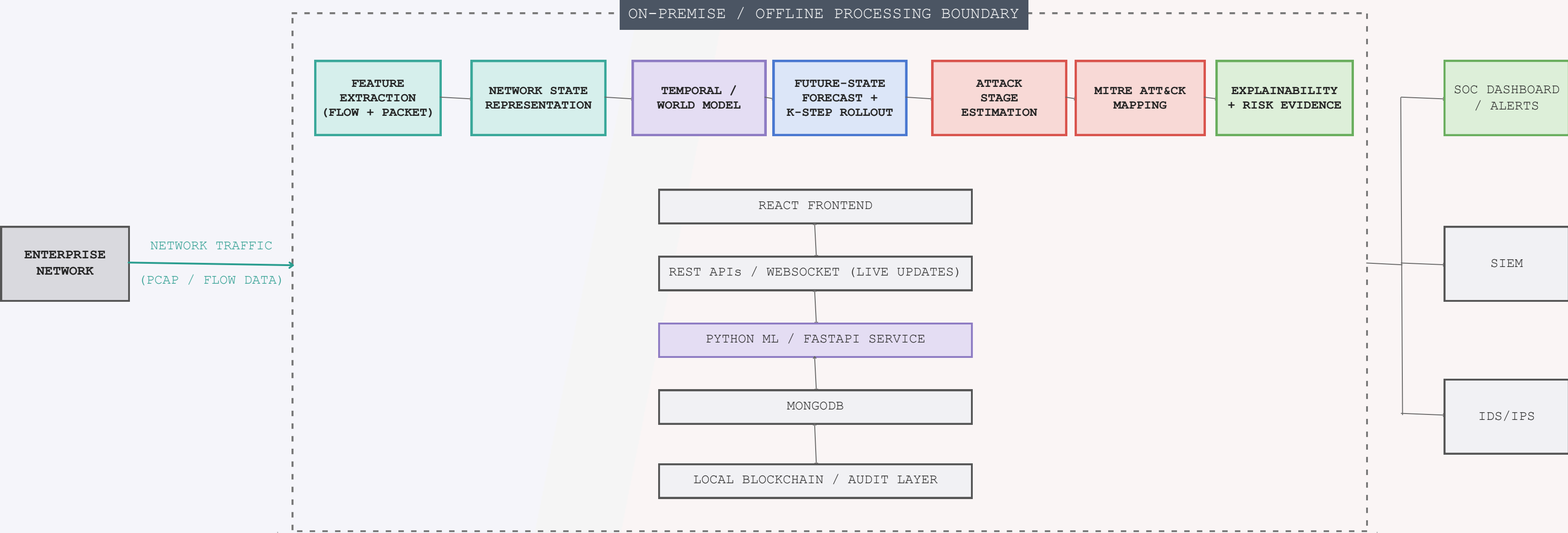


From Detecting the Present to Forecasting the Attack Trajectory

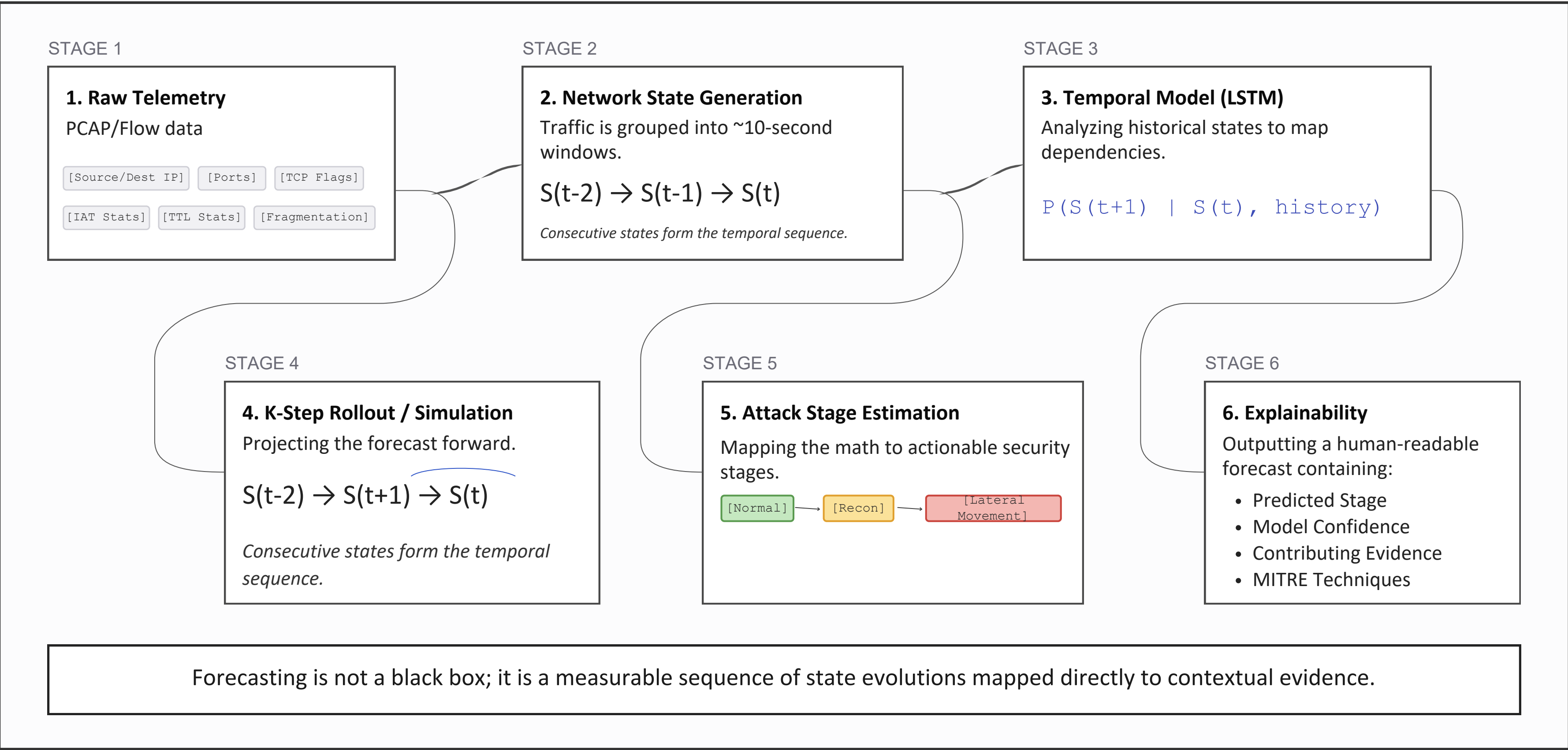


End-to-End Technical Architecture and Data Flow



Sensitive network telemetry can remain entirely within the organization's infrastructure instead of being sent to external cloud services.

How Network Traffic Becomes a Future-State Forecast



Implemented Prototype — From PCAP to Cyber Intelligence

Technology Stack & Data	Database Entities & Active Modules	Verification Checklist (Current Implementation)
 Frontend: React  Backend: Python, FastAPI, REST APIs, WebSocket/Socket.IO (Live Updates)  ML Engine: LSTM / Temporal Forecasting  Database: MongoDB  Audit: Local Blockchain  Current Demo Data: CTU-13 Neris Botnet PCAP	Entities: • Hosts • Traffic Events • Forecasts • Alerts • Audit Logs Modules: • Network Status • Threats • Live Forecast • Attack Progression • Network Graph • Investigation • Explainability • Risk Intelligence	✓ PCAP/network traffic analysis ✓ 10-second network-state generation ✓ Temporal forecasting & attack-stage estimation ✓ MITRE ATT&CK mapping ✓ Forecast history & live WebSocket updates ✓ Explainability/evidence generation ✓ Local blockchain audit verification ✓ Complete local/offline deployment

A functioning, deployable prototype verified with CTU-13 botnet data, moving beyond theoretical models to active intelligence.

Beyond Attack Forecasting — Toward Security & Quantum Readiness

Current System Differentiation

Traditional Detection

Traffic → Current Anomaly → Alert → Investigation

Focuses on isolated events.

NETRA AI

Traffic → Network-state evolution → Future-state forecast → Explainable warning

Trajectory-aware, K-step rollout, strictly on-premise.

FUTURE ENHANCEMENT

Quantum Threat Readiness / HNDL (Harvest Now, Decrypt Later) Exposure

Estimating future exposure to quantum-enabled data decryption.

$$QTS = 0.30C + 0.20S + 0.20L + 0.15P + 0.15H$$

C: Cryptographic Vulnerability (e.g., RSA/ECC = higher exposure)

S: Data Sensitivity (Financial/Customer data)

L: Data Lifetime (Long-lived data = higher HNDL concern)

P: PQC Readiness (No post-quantum cryptography migration yet)

H: HNDL-like Activity (Suspicious access to encrypted data)

**Note: This is a custom project risk metric, not an official NIST standard.*

From detecting the present to understanding the trajectory — and preparing the enterprise for future cryptographic risk.